Hwang 2022 PDAC — analysis summary

RUN 4 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Pre-Harmony the 29 tumours form a per-patient archipelago (iLISI 1.0 of 29; same-sample neighbour frac 0.95). Harmony on sample intermixes them into one continuous mass — iLISI 8.8, same-sample 0.17, a single kNN-graph component after complete island removal (107 detached-speck nuclei → 63,997). θ=2 from the sweep (mixing saturates by θ=2; geography θ-invariant). Three density lobes, all 2–8% cycling — no proliferation compartment.

Gate 2 — module geography → NO-GO

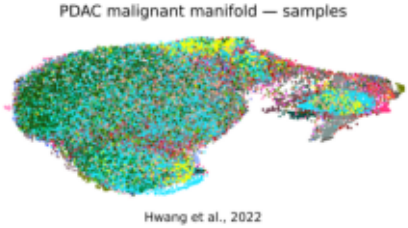
Judged on geography vs an explicit null, not the means (flattest in the atlas: Gast −0.004 / pSom −0.002 / Crest +0.003). The embryonic modules carry NO coherent territory: each module's Moran's I sits at/below its own size-matched union-pool null (Gast I=0.06 z=−0.4 p=0.65; pSom z=−0.2 p=0.55; Crest z=−0.5 p=0.59). Gast↔pSom spatial r=−0.15, far short of the passing same-tissue PDAC bar (Peng/Steele/Moncada/Lin, −0.53 to −0.68); no terminal Crest (module-high deciles mid-pseudotime). The manifold is organized by PDAC meta-programs (EMT/Hypoxia/Chromatin), not the developmental axis → the embryonic pattern is absent.

Gate 3 — cross-repression → NA

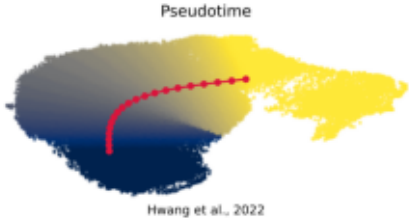
Not applicable — Gate 2 did not pass, so there is no Gast/pSom hub and no bifurcation to test. The branching-only 04_correlation / 05_modules notebooks are not produced (§6); the EMT-TF step is the final notebook (04).

STEP FIGURES

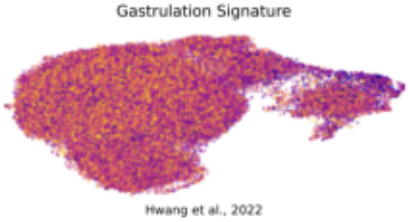
Manifold — samples



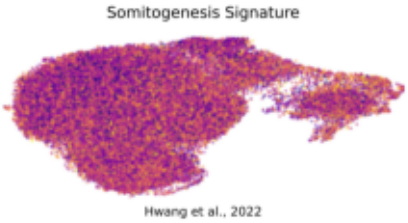
Pseudotime / curve



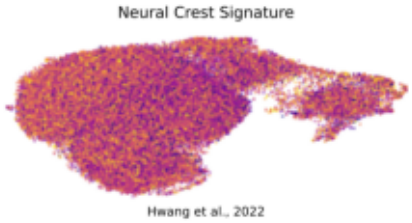
Gastrulation



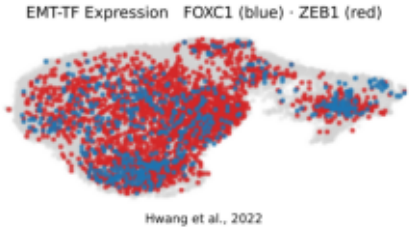
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

PDAC (Hwang et al., 2022), 10x snRNA-seq — 43 tumours / 224,988 cells (master scope); malignant only: 64,538 nuclei → 63,997 across 29 samples (≥100-cell floor; the multiplicative count gap is degenerate, sitting in the speck tail). Whole malignant manifold; three density lobes, not scoped.

Verdict.

The embryonic pattern does NOT reproduce. Gate 1 passes — one continuous, sample-integrated malignant manifold — but the developmental modules have no coherent geography (all at/below random-null level), no Gast↔pSom cross-repression axis (spatial r=−0.15 vs passing PDAC −0.53..−0.68), and no terminal Crest. The manifold is structured by PDAC meta-programs, not gastrulation/somitogenesis. EMT-TFs FOXC1 (blue) and ZEB1 (red) mark the manifold descriptively; canonical SNAI1/TWIST are below the detection floor. G1 GO / G2 NO-GO / G3 NA. Stopped at Gate 2 per §6; 01–03 + expression(04) produced.

Parameter choices

Sample threshold	≥100-cell floor (gap 2.67× in speck tail → degenerate); 29/43 kept	**Overdispersed genes**	1,645 (scFates find_overdispersed); no sc.pp.scale
Integration	Harmony on sample, θ=2 (iLISI 1.0→8.8; geography θ-invariant)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 107 specks removed; 1 kNN comp
Geometry	Non-trajectory (Gate 2 NO-GO); probe curve 2 tips/0 forks, no hub	**Module means (raw)**	Gast −0.004 · pSom −0.002 · Crest +0.003 (geography decides)
Root	curve tip 0 (max Gast+pSom−Crest); ordering does not reproduce	**EMT-TFs shown**	FOXC1 (3.6%) · ZEB1 (15.7%); SNAI1/TWIST <0.1%; VIM excluded